

Dr. Zahir Ahmad Shah

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Research Interests

My research interest mainly involves the study of **broadband spectral and temporal properties of blazars** using multi wavelength data from various astrophysical observatories. This includes **developing analysis scripts** to automate the data reduction process and numerical codes to interpret the reduced data using theoretical emission models. I am also involved in obtaining the **analytical/numerical solutions of the particle distribution in the acceleration and cooling regions** for different scenarios of perturbations to the acceleration time-scale, cooling time-scale and injection rates. Besides this, I am also involved in characterization of temporal properties of longterm light curves of blazars.

Education

Ph.D.	University of Kashmir	2018
<i>Thesis: Understanding the Spectral and Temporal Behaviour of High Energy Blazars</i>		
<i>Supervisor: Prof. Naseer Iqbal</i>		
<i>Co-Supervisor: Dr. Sunder Sahayanathan</i>		
M.Phil.	University of Kashmir	2014
M.Sc. in Physics	University of Kashmir	2011
B.Sc.	University of Kashmir	2009
Higher Secondary	J&K State Board of School Education	2006
Secondary School	J&K State Board of School Education	2004

National Exam

› Qualified Joint CSIR-UGC National Eligibility Test (NET), All India Rank: 15.

Academic Positions

2018–2021	Post-Doctoral Fellow, Inter-University Centre for Astronomy and Astrophysics
2021–14 June 2026	INSPIRE Faculty, Central University of Kashmir
15 June 2026–Present	Assistant Professor & INSPIRE Faculty, Manipal Centre for Natural Sciences,

Teaching Experience at University Level

- › Classical Mechanics
- › Quantum Mechanics
- › Relativistic Quantum Mechanics
- › Quantum Field Theory

- › Mathematical Physics
- › High Energy Astrophysics

Research Grants

- › Co-investigator, ISRO AstroSat data utilization project, Rs. 18.37 Lacs.
- › DST INSPIRE Faculty Fellowship of 77.40 lacs and research grant of Rs. 35 Lacs over five years.

PROFESSIONAL SERVICE

- › Served as a Peer-reviewer for the Nature Journals (**Nature Communications**.)
- › Served as a Peer-reviewer for the **The Astrophysical Journal Supplement Series**.
- › Served as a Peer-reviewer for the **Monthly Notices of the Royal Astronomical Society**.

Research Highlights

- › **Broadband spectral and temporal data analysis techniques** across multiple wavelengths.
- › **Theoretical spectral modeling** of blazars.
- › **Time resolved spectral study** of blazars.
- › **Constraining High Energy (X-ray and gamma-ray) emission** in Blazars.
- › **Prediction of VHE Blazars** for upcoming VHE telescopes.
- › **Obtain constraints on EBL estimation** using the VHE observations.
- › **Numerical/Analytical solutions** of Fokker-Plank equations
- › **Statistical and simulation study** of blazar light curves.

Research Expertise

- › **Data analysis:** Fermi-LAT, TACTIC, Swift-XRT, Swift-UVOT, NuStar, AstroSat SXT/LAXPC, UVIT, .
- › **Theoretical SED modeling, numerical/analytical solutions** for Fokker-Plank equation.
- › **Computational/graphical skills:** Python, FORTRAN, XSPEC, PYXSPEC, Gnuplot, Matplotlib.

Research Fellowships

- › Junior and Senior Research Fellowship (JRF & SRF), ISRO-RESPOND Project.
- › Post-Doctoral Fellowship, IUCAA.
- › INSPIRE Faculty Fellowship, DST.

Collaborations

- › University of Kashmir, ApSD BARC Mumbai, IUCAA Pune, IIA, IISER Mohali, Tezpur University, University of Southampton, NIT Rourkela, University of Nizwa

Accepted Proposals

- *Principle Investigator (PI)* in “Investigating the Spectral Breaks and Thermal Emission of High Red shift Flat Spectrum Radio Quasars using AstroSat”, A03-77
- *Principle Investigator (PI)* in “Investigating the broad band Spectral energy distribution Bl Lac blazars using AstroSat”, A04-094
- *Principle Investigator (PI)* in “Multi-wavelength observation of FSRQ PMNJ1923-2104 during its active phase using AstroSat”, T03-018
- *Co-Investigator (Co-PI)* in “Observation of Mrk 421 with AstroSat during the present flaring activity”, T01-218
- *Co-Investigator (Co-PI)* in “Investigation the broadband spectral shape of Mrk 180 using multi-wavelength observation from AstroSat”, A05-064
- *Co-Investigator (Co-PI)* in “X-ray/UV study of the high energy peaked blazar PKS 0352-686”, A05-202

Publications

1. Malik Zahoor, Akbar Sikandar, **Shah Zahir**, van Soelen B., “Exploring the Transitional Parameter Space of Blazars using Gamma-ray and X-ray Population Diagnostics”, *JHEAP*, Volume 56, February 2027, 100753, <https://doi.org/10.1016/j.jheap.2026.100753> (**Impact Factor-6.1**)
2. Peer Anjum, **Shah Zahir**, Akbar Sikandar, Maqbool Bari, Misra Ranjeev, “Probing the High-Energy Emission of the VHE-emitting Changing-Look Blazar B2 1420+32”, *ApJ*, 2026, accepted for publication, arXiv:2607.20375 (**Impact Factor-5.7**)
3. Akbar Sikandar, **Shah Zahir** et al, “A Possible Short-Timescale Optical Quasi-Periodic Oscillation in PKS 0805-07 from High-Cadence TESS Observations”, *JHEAP*, Volume 54, August 2026, 100693.
4. Manzoor Aaqib, **Shah Zahir** et al “Broad-band spectral and temporal study of OP 313 during its 2025 January very high energy flare”, *JHEAP*, Volume 53, July 2026, 100619 (**Impact Factor-6.1**)
5. Ahanger Sajad, **Shah Zahir** et al “Modeling the Time-variable Broadband Emission and Correlation Study of FSRQ S5 1044+71”, 2026, *ApJ*, 999, 170 (**Impact Factor-5.7**)
6. Nazir Zeeshan, Akbar Sikandar, **Shah Zahir** et al “Broadband Variability Analysis of FSRQ PKS 0402-362 with Indications of Quasi-Periodic Modulation”, 2026, *ApJ*, 998, 227 (**Impact Factor-5.7**)
7. Akbar Sikandar, **Shah Zahir** et al “Spectral state switching in Mrk421: Results from the AstroSat LAXPC/SXT observations”, *JHEAP*, Volume 53, July 2026, 100633 (**Impact Factor-6.1**)
8. **Shah Zahir**, et al, “Comprehensive variability analysis of blazars using Fermi light curves across multiple timescale”, *Phys. Rev. D*, 2025, 111, 123052 DOI: <https://doi.org/10.1103/61tz-jk8c> (**Impact Factor-5.3**)
9. Peer Anjum, Dar Athar A, **Shah Zahir** et al “Multiwavelength spectral and temporal analysis of VHE Blazar 1ES 1959+650: Tracing emission mechanisms across flux states”, *MNRAS*, 2026, 545, 3, staf2031, <https://doi.org/10.1093/mnras/staf2031> (**Impact Factor-5.2**)

10. Sikandar A., **Shah Zahir**, et al., “Indication for dual periodic signatures in PKS 0805-07 from multi-technique time series analysis”, *Phys. Rev. D (PRD)*, 2025, 112, 063061, <https://doi.org/10.1103/physrevd.112.063061> (**Impact Factor-5.3**)
11. Dar Athar, **Shah Zahir**, et al, “Unraveling the broadband spectrum of B2 1308+326: A clue to its high energy emission mechanism”, *New Astronomy*, 121 (2025) 102463, <https://doi.org/10.1016/j.newast.2025.102463> (**Impact Factor-2.1**)
12. **Shah Zahir**, “Multi-wavelength variability and broadband SED modeling of BL Lac during a bright flaring period MJD 59000-59943”, *MNRAS*, 2024, 527, 5140–5154 (**Impact Factor-5.2**)
13. Malik Zahoor, Akbar Sikandar, **Shah Zahir**, et al, “Statistical Insights into Flux and Photon Index Distributions of VHE FSRQs from Fermi-LAT Observations”, *MNRAS*, 2025, 539, 3, 2185–2201, <https://doi.org/10.1093/mnras/staf620> (**Impact Factor-5.2**)
14. Tantry Javaid, Sharma Ajay, **Shah Zahir**, Iqbal Naseer and Bose Debanjan “Study of multi-wavelength variability, emission mechanism and quasi-periodic oscillation for transition blazar S5 1803+784”, *JHEAP*, Volume 47, July 2025, 100372 (**Impact Factor-6.1**)
15. Ahanger Sajad, Sahayanathan Sunder, Sitha K. Jagan, **Shah Zahir**, Naseer Iqbal “Multi-wavelength analysis of FSRQ B2 1348+30B: Constraints on the jet power”, *JHEAP*, 2025, 47, 100400, <https://doi.org/10.1016/j.jheap.2025.100400> (**Impact Factor-6.1**)
16. Sikandar A., **Shah Zahir**, et al., “Probing spectral evolution and intrinsic variability of Mkn 421: A multi-epoch AstroSat study of X-ray spectra”, *JHEAP*, 2025, 45, 438-455 (**Impact Factor-6.1**)
17. **Shah Zahir**, et al. “AstroSat observation of the HBL 1ES 1959+650 during its October 2017 flaring”, *MNRAS*, 2021, 504, 5485–5495 (**Impact Factor-5.2**)
18. **Shah Zahir**, et al. “Unveiling the broadband spectral and temporal properties of PKS 0903-57 during its brightest flare”, *MNRAS*, 2021, 504, 416–427 (**Impact Factor-5.2**)
19. Tantry J., **Shah Zahir** et al. “Probing broadband spectral energy distribution and variability of Mrk 501 in the low flux state”, *JHEAP*, 2024, 44, 393–409 (**Impact Factor-6.1**)
20. Sikandar A., **Shah Zahir**, et al. “Insights into the Long-Term Flaring Events of Blazar PKS 0805-07: A Multi-Wavelength Analysis over the 2009-2023 Period”, *The Astrophysical Journal (ApJ)*, 2024, 977, 111 (**Impact Factor-5.4**)
21. **Shah Zahir**, et al. “On the determination of log-normal flux distributions for astrophysical systems”, *MNRAS*, Volume 496, Issue 3, August 2020, Pages 3348–3357 (**Impact Factor-5.2**)
22. **Shah Zahir**, et al. “Study on temporal and spectral behavior of 3C 279 during 2018 January flare”, *MNRAS*, 484, 3168-3179 (2019) (**Impact Factor-5.7**)
23. Manzoor A., **Shah Zahir**, et al. “Broadband spectral and temporal study of Ton 599 during the brightest January 2023 flare”, *MNRAS*, 2024 (**Impact Factor-5.2**)
24. **Shah Zahir**, et al. “Log-normal flux distribution of bright Fermi blazars”, *RAA*, 2018, 18, 11, 141 (**Impact Factor-2.8**)
25. **Shah Zahir**, et al. “Clues on high-energy emission mechanism from blazar 3C 454.3 during 2015 August flare”, *MNRAS*, 2017, Volume 470, Issue 3, p.3283-3299 (**Impact Factor-5.2**)
26. Khatoon R, **Shah Zahir**, et al “Study of long-term flux and photon Index distributions of blazars using RXTE observations”, *MNRAS*, 2020, 491, 2, 1934–1940 (**Impact Factor-5.2**)

27. Khatoon R, **Shah Zahir**, et al., “Correlations between X-ray spectral parameters of Mkn 421 using long-term Swift-XRT data”, *MNRAS*, 2022, 515, 3749–3759 (**Impact Factor-5.2**)
28. Rajput B., **Shah Zahir**, et al. “Correlation between optical and γ -ray flux variations in BL Lacs”, *MNRAS*, 2021, 504, 1772–1786 (**Impact Factor-5.2**)
29. Deka K., **Shah Zahir**, et al “The long-term X-ray flux distribution of Cygnus X-1 using RXTE-ASM and MAXI observations”, *JHEAP*, 31, 2021, 23-30 (**Impact Factor-6.1**)
30. Hota J., **Shah Zahir**, et al “Understanding the X-ray spectral curvature of MKN 421 using broadband AstroSat observations”, *MNRAS*, 2021, 508, 5921–5934 (**Impact Factor-5.2**)
31. M Zahoor, **Shah Zahir**, et al “Multiwavelength study of blazar 4C + 01.02 during its long-term flaring activity in 2014-2017”, *MNRAS*, 2022, 514, 4259-4269 (**Impact Factor-5.2**)
32. Dar A., **Shah Zahir**, et al “Can FSRQ 3C 345 be a very high energy blazar candidate?”, *MNRAS*, 2024, 527, 10575-10583 (**Impact Factor-5.2**)
33. M Zahoor, Sahayanathan, **Shah Zahir**, et al “Model-independent Redshift Estimation of BL Lac Objects through VHE Observations”, *MNRAS*, 2022, 511, 994–1003 (**Impact Factor-5.2**)
34. M Zahoor, Sahayanathan, **Shah Zahir**, et al “Very-high-energy flat spectral radio quasar candidates”, *MNRAS*, 2022, 515, 3, 4505-4513 (**Impact Factor-5.2**)
35. Xiongfei G., ..., **Shah Zahir**, et al “Exploring γ -Ray Flares in the Long-term Light Curves of CTA 102 at GeV Energies”, *The Astrophysical Journal Supplement Series*, 2022, 260:48 (29pp) (**Impact Factor-8.8**)
36. Manzoor A., **Shah Zahir**, et al “Understanding the very high energy γ -ray excess in nearby blazars using leptonic model”, *MNRAS*, 2023, 525, 3, 3533-3540 (**Impact Factor-5.2**)
37. Thekkoth A., Sahayanathan, **Shah Zahir**, et al “Understanding the broad-band emission process of 3C 279 through long term spectral analysis”, *MNRAS*, 2023, 526, 4, 6364-6380 (**Impact Factor-5.2**)
38. Khatoon R., Prince R, **Shah Zahir**, et al “Temporal and spectral study of PKS 0208-512 during the 2019–2020 flare”, *MNRAS*, 2022, 513, 611-623 (**Impact Factor-5.2**)
39. M Showkat, I Naseer, **Shah Zahir**, “Search of Astrophysical results with LIGO from the Science runs S1 to S5”, *JOURNAL OF ADVANCES IN PHYSICS* 6 (3)
40. Khan S, H. Abdullah; **Shah Zahir**, et al “The Fluctuation Theory, Critical Phenomena and Gravitational Clustering of Galaxies”, 2025, *Journal of Astrophysics and Astronomy*, 46, 2, 75 (**Impact Factor-1.6**)

ArXiv Preprints (Under Review):

41. Dar Athar A., **Shah Zahir** et al, “Signature of hadronic emission in gamma-ray spectrum of B2 1308+326”, <https://doi.org/10.48550/arXiv.2503.01583>
42. **Shah Zahir**, Akbar Sikandar, “Advance warning of γ -ray blazar flares from *Fermi*-LAT light curves: a strictly causal machine-learning backtest”, 2026, arXiv:2605.07680
43. Malik Zahoor, Akbar Sikandar, **Shah Zahir**, Rubab Seemin, “Indication of γ -ray Quasi-periodicity in GB6 J1037+5711 from Multi-technique Timing Analysis”, 2026, arXiv:2604.27461

44. Akbar Sikandar, **Shah Zahir**, “Characterizing Short-Timescale Optical Variability in Non-blazar Active Galactic Nucleus PKS0521–36 Using TESS”, 2026, arXiv:2603.22197

Conference Proceedings:

45. Khatoon R., **Shah Zahir** et al. “Flux Distribution Study of Mkn 421 with SPOL, RXTE and Fermi-LAT Telescopes, *Springer Proceedings in Physics* (Selected Progresses in Modern Physics: Proceedings of TiMP 2021), 2022, 125–133, DOI:10.1007/978-981-16-5141-0_13.
46. Deka K., **Shah Zahir** et al. “A study of the effects of histogram binning on the accuracy of finding flux distribution of X-ray binaries”, *Materials Today: Proceedings*, Volume 65, Part 5, 2022, Pages 2862-2864
47. Deka K., **Shah Zahir** et al. “Distribution of X-ray flux: RXTE-PCA Observation of Cygnus X-1”, *Materials Today: Proceedings*, Volume 65, Part 5, 2022, Pages 2862-2864.
48. Deka K., **Shah Zahir** et al. “Distribution of X-ray flux: RXTE-PCA observation of Cygnus X-1, Conference: Springer International Conference on Trends in Modern Physics 26-27 February 2021.
49. **Shah Zahir** et al “Understanding the evolution of X-ray spectral properties of blazars using broad-band AstroSat observations”, in 44th COSPAR Scientific Assembly, vol. 44, Jul. 2022, p. 2243.
50. J. Hota, **Shah Zahir** et al. “X-ray Spectral study of MKN 421 using Astrosat observation, in 44th COSPAR Scientific Assembly, vol. 44, Jul. 2022, p. 2245.
51. R. Khatoon, R. Misra, **Zahir Shah**, J. Hota, R. Gogoi, and A. Charan Pradhan, “Correlations between X-ray spectral parameters of Mkn 421 using long-term Swift-XRT data,” in 44th COSPAR Scientific Assembly, vol. 44, Jul. 2022, p. 2075.
52. Akbar Sikandar, **Shah Zahir**, Misra R., Iqbal N., Tantry J., “Long term correlation study between X-ray spectral parameters of Mkn 421 using AstroSat data”, 42nd Meeting of the Astronomical Society of India (ASI), 2024, P149.
53. Khatoon R., Jithesh V., Prince R., **Shah Zahir**, Misra R., Gogoi R., “Swift follow-up of the renewed flaring phase of blazar SBS 1150+497”, *The Astronomer’s Telegram*, 2019, ATel #13285.

Conference Presentations

➤ Oral Presentations/Resource person:

- International conference on *A Decade of AstroSat Observations: Science Outcomes and Future Prospects*, 30 January –1 February 2026, delivered contributed talk titled “Time resolved X-ray spectral study of Blazars with AstroSat: Probing underlying particle distribution”.
- Delivered IIA Astrophysics Seminar at Indian Institute of Astrophysics titled “Broadband Views of Blazars: Spectral and Temporal Probes of Relativistic Jets” on 27th January 2026
- *Astrosat & XPoSat Workshop 2024*, 28 February – 03 March 2024 at Providence Womens College Kozhikode, participated as resource person in the Workshop.
- *Advanced Astrosat Data Analysis Workshop 9-15 January 2023*, participated as resource person in the Workshop.

- *Himalayan Meet of Astronomers 2023 at IUST, Awantipora, 25-26 September 2023*, gave an invited talk on “Understanding Broadband Emission from Blazars” and chaired one scientific session on blazars.
- *4th National Conference on High Energy Emission from Active Galactic Nuclei*, gave an invited talk on “Temporal behavior of blazar light curves” and chaired one scientific session on blazars.
- *Introductory Workshop in Astronomy and Astrophysics, 18–19 December, 2021*, gave an invited talk on “Active Galaxies - Brightest Objects in the Universe”.
- *2nd China-India Workshop on High Energy Astrophysics, 6–10 December 2021*, gave an invited talk on “Understanding the evolution of X-ray spectral properties of blazars using broadband AstroSat observations”.
- *IUCAA Annual Seminars by Post-Doctoral Fellows - 2021*, gave a presentation on “Broadband Spectral and Temporal Study of Blazars”.
- *One day online meeting on Theoretical Aspects of High Energy Emission from AGN held on 06 July 2021*, gave an invited talk on “Time Resolved Spectral Study of 1ES,1959+650 using AstroSat Observations.
- *Astrophysical jets and observational facilities: National perspective, 05-09 April 2021*, gave a talk on “AstroSat observation of the HBL 1ES,1959+650 during its October 2017 flaring”.
- *Astronomy Group Seminar Programme, University of Southampton*, gave a seminar “Flux distribution study of Fermi bright blazars and simulated light curves” on Mon 14 Oct 2019.
- *4th National conference on Recent Trends in the Study of Compact Objects: Theory and Observation (RETCO-IV) 17 - 20 April 2019, IUCAA, Pune*, presented the work “Study on temporal and spectral behaviour of 3C 279 during 2018 January flare”
- *Growth Points in Physics (GPP-2017)*, UGC Sponsored National Seminar, held at University of Kashmir, Srinagar, 21-23 October 2017
- *High Energy Emission from Active Galactic Nuclei, 3rd national conference*, held at University of Calicut, Kerala, 28-30 November, 2017
- *Last Friday Talk at IUCAA on 25 January 2019*, presented the work “Multiwavelength study of 3C 279 during 2018 January flare”
- *Astronomical Society of India (ASI), 34th meeting* held at Kashmir University, Srinagar, 10-13 May, 2016

➤ Poster Presentations:

- *Young Astronomers Meet (YAM) 2017*, organized by the IUCAA during 11-15 September, 2017
- *IUCAA Science day foundation 2019*, organized by the IUCAA on 28 February, 2019

➤ Participations:

- *AstroSat Calibration Meeting*, Hosted by AstroSat Science Support Cell, IUCAA, Pune , 23-24 August 2022.
- *AstroSat Data analysis*, workshop organised by IUCAA, 13-16 Nov. 2017

- *AstroSat AO Proposal submission*, workshop organised by IUCAA, 13-14 July 2016
- *Extragalactic Relativistic Jets: Cause and Effect*, hosted by ICTS Bangalore, 12-20 Oct. 2015
- *Radio Astronomy School*, organized by National Center for Radio Astronomy (NCRA), Pune, 19-30 August 2013
- *Astronomical Techniques and Science with Virtual Observatories*, jointly organized by Department of Physics (Kashmir University) and Inter University Centre for Astronomy and Astrophysics (IUCAA, Pune), 23-26 Sep. 2013
- *High Energy Emission from Active galactic Nuclei (AGN)*, Jointly organised by department of physics (University of Kashmir), Board of Research in Nuclear Science (BRNS) BARC., and Indian Space Research Organisation (ISRO)., 7-9 Oct. 2013
- *Stellar Astrophysics*, jointly organized by Department of Physics (University of Kashmir, Srinagar) and Inter University Centre for Astronomy and Astrophysics (IUCAA), Pune, 24-26 Oct. 2011
- *High Energy Astrophysics*, winter School, organized by ApSD Bhaba Atomic Research Centre (BARC) at Mount Abu Rajasthan, 28 January to 3rd February, 2011
- *Participate in Online International workshop on VHE Data Analysis using Open Source Packages*, University of Kashmir, Srinagar, 1-3 February 2022.

References

- > **Prof. Naseer Iqbal**, Department of Physics, University of Kashmir, dni_phtr@kashmiruniversity.ac.in
- > **Prof. Ranjeev Misra**, IUCAA PUNE, rmisra@iucaa.in
- > **Prof. Debanjan Bose**, Department of Physics, Central University of Kashmir, Ganderbal, debanjan.bose@cukashmir.ac.in